

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

SUMMER SEMESTER, 2020-2021

DURATION: 3 HOURS

FULL MARKS: 150

Math 4641: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Marks of each question and corresponding CO and PO are written in the right margin.

1. a) Use Taylor's polynomial to derive the orders of error for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for choosing one of these three. 9+5
(CO1, CO3)
(PO1)
- b) A tissue paper company advertises that every roll of their toilet paper has at least 250 sheets. The probability that there are 250 or more sheets in the toilet paper roll is given by the following equation: 11
(CO2)
(PO2)

$$P(y \geq 250) = \int_{250}^{\infty} 0.3515e^{-0.3881(y-252.2)^2} dy$$

By theoretical reasoning, the above integral calculates the probability up to infinite sheets. But practically, it has been seen that the number of sheets rarely exceed 270. So, considering 270 as the maximum possible number of sheets, the above integral can be approximated as follows:

$$P(y \geq 250) = \int_{250}^{270} 0.3515e^{-0.3881(y-252.2)^2} dy$$

Use the 2-segment Trapezoidal rule to find the probability that there are 250 or more sheets in a random toilet paper roll.

2. a) Without using linearization transform, describe the individual solution approaches for the following nonlinear regression models: 4+5+4
(CO1)
(PO1)
 - i. Exponential model
 - ii. Growth model
 - iii. Polynomial model
- b) What is the drawback in directly using the first 3 terms of the Taylor series for deriving y_{i+1} in Runge-Kutta 2nd order method? What is the alternative way of approximating the first 3 terms of the series? 3+3
(CO3)
(PO2)
- c) What are the different forms of the approximation of Runge-Kutta 2nd order method based on Heun's method, Midpoint method and Ralston's method? 6
(CO1)
(PO1)
3. a) Find the Eigenvalues and Eigenvectors of the following matrix A: 12
(CO2)
(PO1)

$$A = \begin{bmatrix} 2 & 27 & 0 \\ 0 & 4 & 40 \\ 0 & 3 & 30 \end{bmatrix}$$
- b) Describe how linearization transform is applied on the following nonlinear models: 2+2+2
(CO1)
(PO1)
 - i. Exponential model
 - ii. Power model
 - iii. Growth model
- c) Why do we normally use polynomials as interpolating functions? Can single polynomials always be used to accurately interpolate data? 2+5
(CO3)
(PO2)

4. a) For approximating $I = \int_a^b f(x)dx$ using Simpson's 3/8 rule, how are the points chosen over the interval $[a, b]$, and what are their exact coordinates in terms of a and b ? Provide necessary illustrations. (CO1) (PO1) 9
- b) The minimum number of segments for Simpson's 3/8 rule is 3. Given that the 3-segment formula is $I = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$, derive the general n-segment formula? (CO1) (PO2) 8
- c) What is the LU decomposition of A , where $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$? (CO2) (PO1) 8
5. a) What is the motivation behind the use of LU decomposition? Provide a scenario where the use of LU decomposition can enhance the performance of a numerical analysis system. (CO3) (PO2) 5+5
- b) Given n data pairs $(x_1, y_1), \dots, (x_n, y_n)$, do the values of the constants a_0 and a_1 in the least square regression line model $y = a_0 + a_1x$ correspond to the absolute minimum of the sum of the squares of residuals? Are the constants of regression unique? (CO1) (PO2) 9+6
6. a) To maximize the catch of bass in a lake, it is suggested to throw the line to the depth of the thermocline. You can infer that there is a thermocline at a particular depth by noticing the sudden change in temperature. The following table provides us the depth vs temperature data for the lake: (CO2, CO3) (PO3) 6+7+3

Table 1: Depth vs Temperature

Depth, $h(m)$	Temperature, $T(^{\circ}C)$
0	19.1
-1	19.1
-2	19
-3	18.8
-4	18.7
-5	18.3
-6	18.2
-7	17.6
-8	11.7
-9	9.9
-10	9.1

It is noticed that there is a large difference in temperature from $h = -7$ to $h = -8$, so the thermocline must be somewhere between these two depths. The intermediate temperature data between these two depths is missing.

Determine the temperature at $h = -7.5$ using a **first order Lagrange polynomial** and then a **second order Lagrange polynomial** and find the absolute relative approximate error.

- b) Derive Simpson's 1/3 rule using the method of coefficients. Using necessary illustrations, determine the geometric interpretation of this derivation? (CO1) (PO1) 5+4